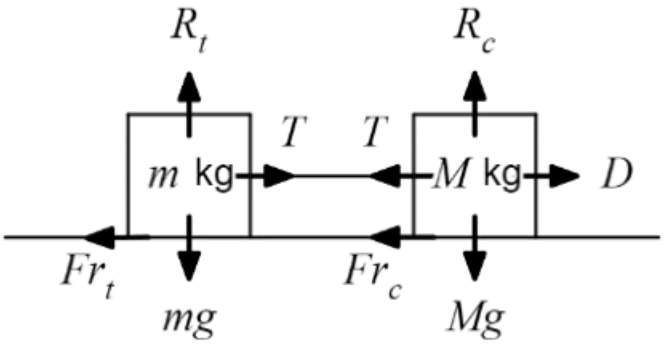
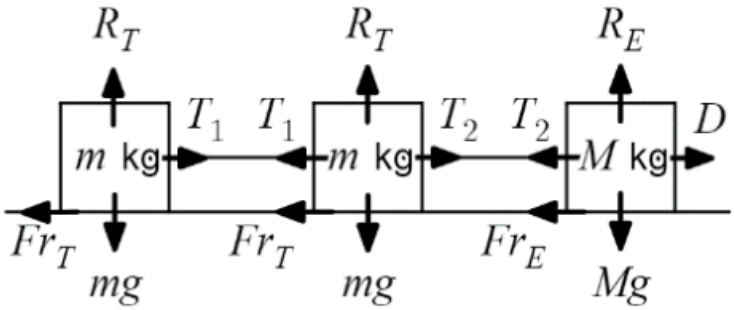
1.69 Forces: Connected Particles

Worked Solutions

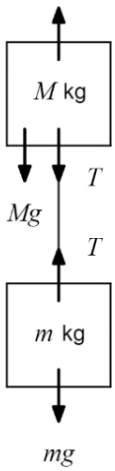
1 a)



1 b)



1 c)



2 a)

On both R($\rightarrow$): $600 = 2500a \Rightarrow a = 0.24$ ms^{-2}

2 b)

On car R($\rightarrow$): $600 - T = 2000 \times 0.24 \Rightarrow T = 120$ N

3 a)

Same acceleration throughout.

On both R($\rightarrow$) : $1100 - 72 - 16 = 2600a$

$$a = \frac{253}{650} = 0.389 \text{ ms}^{-2} \text{ (3 SF)}$$

3 b)

On trailer R($\rightarrow$) : $T - 16 = 400 \times \frac{253}{650} \Rightarrow T = \frac{2232}{13} = 172$ N (3 SF)

4 a)

On both R($\rightarrow$): $60 - 12 - 3 = 7.5a \Rightarrow a = 6$ ms^{-2}

4 b)

On Q R($\rightarrow$): $T - 3 = 1.5 \times 6 \Rightarrow T = 12$ N

4 c)

Inextensible: the acceleration of the two particles is the same Light: the mass is negligible.

5)

Let k be the proportional constant Resistance against P: $10k$ Resistance against Q: $2k$ On both R($\rightarrow$): $30 - 12k = 12 \times 0.5$ so $k = 2$ On P R($\rightarrow$): $30 - 20 - T = 10 \times 0.5$ so $T = 5$ N

6 a)

On both R($\rightarrow$): $120 = (10 + m) \times 3$ so $m = 30$

6 b)

On P R($\rightarrow$): $120 - T = 10 \times 3$ so $T = 90$ N

7 a)

Both R($\rightarrow$) : $37.8 - 6 - 2.4 = 7m \times a \Rightarrow a = \frac{4.2}{m}$

7 b)

P R($\rightarrow$): $37.8 - T - 6 = 5 \times 4.2 \Rightarrow T = 10.8$

7 c)

Inextensible: the acceleration of the two particles is the same Light: the mass is negligible.

8 a)

All R($\rightarrow$): 25000 -2400 $-2R=$ 17000 $\times$ 1.2 $R=$ 1100 N

8 b)

On engine: R($\rightarrow$) : $25000 - 2400 - T_1 = 10000 \times 1.2$

So $T_1 = 10\,600$ N

On middle truck:

$$\text{R}(\rightarrow) : 10\,600 - 1100 - T_2 = 3500 \times 1.2$$

So $T_2 = 5300$ N

So the tensions in the towbars are 10 600 N and 5300 N

9)

Both R($\rightarrow$): 300 -180 $-R=$ 0 $\Rightarrow R=$ 120 N

10 a)

$5^2 = 0^2 + 2 \times a \times 20 \Rightarrow a = 0.625$ ms^{-2}

10 b)

On both R($\rightarrow$): $F-90$ $-40 = 2600 \times 0.625 \Rightarrow F=$ 1755 N

10 c)

On car R($\rightarrow$): 1755 $-T-90 = 1800 \times 0.625 \Rightarrow T=$ 540 N

11 a)

Both R($\rightarrow$): 145.6 -60 $-18 = 26 \times a \Rightarrow a=$ 2.6 ms^{-2}

11 b)

$s = 0 + \frac{1}{2} \times 2.6 \times 5^2$ so $s = 32.5$ metres

11 c)

P R($\rightarrow$): 145.6 $-T-60 = 20 \times 2.6 \Rightarrow T=$ 33.6 N

12 a)

On both R($\rightarrow$) : $12600 - 900k = 9000a$

$$a = \frac{12600 - 900k}{9000} = \frac{126 - 9k}{90} = \frac{7}{5} - \frac{k}{10} \text{ as req.}$$

12 b)

On engine R($\rightarrow$):

$$12600 - T - 700k = 7000\left(\frac{7}{5} - \frac{k}{10}\right)$$

$$12600 - T - 700k = 9800 - 700k$$

$$T = 2800 \text{ N}$$

12 c)

$3 = 0 + \left(\frac{7}{5} - \frac{k}{10}\right) \times 5 \Rightarrow k = 8$

So $200k$ is $200 \times 8 = 1600$ N

12 d)

So $700k$ is $700 \times 8 = 5600$ N

13 a)

Both R($\rightarrow$): $-2590\ -150\ -60 = 3500 \times a$ So a= -0.8 ms^{-2} Deceleration is 0.8 ms^{-2}

13 b)

$17.5 = 22.5\ -0.8t$ so t= 6.25 seconds

13 c)

Van R($\rightarrow$): $-2590 + T - 150 = 2500 \times -0.8$ T= 740 N

13 d)

The tow bar is inextensible OR it has no mass OR it is horizontal.

13 e)

$0^2 = 17.5^2 - 2 \times 0.4s \Rightarrow s = 382.8125$

so 382 metres (3 SF)

13 f)

No tension would mean that that it would be a higher overall force leading to an increase in acceleration.

14 a)

Both $R(\uparrow)$: $15 - (1.2 + 0.8) \times 9.8 = (1.2 + 0.8) \times a$ a= -2.3 ms^{-2}

14 b)

On Q $R(\uparrow)$: $T - 0.8 \times 9.8 = 0.8 \times -2.3$ so T= 6 N

14 c)

Constant velocity $\Rightarrow a$= 0 Both $R(\uparrow)$: $F - (1.2 + 0.8) \times 9.8 = (1.2 + 0.8) \times 0$ F= 19.6 So an increase of 4.6 N

15 a)

$R(\uparrow)$: $F - 10 \times 9.8 = 10 \times 0.5 \Rightarrow F$= 103 N $\therefore$ By N3L: 103 N 100 N (2 SF – AQA)

15 b)

$R(\uparrow)$: $F - 0.1 \times 9.8 = 0.1 \times 0.7 \Rightarrow F$= 1.05 N $\therefore$ By N3L: 1.05 N 1.1 N (2 SF – AQA)

15 c)

$R(\uparrow)$: $F - 2 \times 9.8 = 2 \times -1.2 \Rightarrow F = 17.2$ N $\therefore$ By N3L: 17.2 N 17 N (2 SF – AQA)

15 d)

$R(\uparrow)$: $F - 0.45 \times 9.8 = 0.45 \times -0.8 \Rightarrow F = 4.05$ N $\therefore$ By N3L: 4.05 N 4.1 N (2 SF – AQA)

16 a)

Both $R(\uparrow)$: $T - 900 \times 9.8 = 900 \times -0.5$ $T = 8370$ N

16 b)

Both $R(\uparrow)$: $T - 900 \times 9.8 = 900 \times 0.5$ $T = 9270$ N

16 c)

Both $R(\uparrow)$: $T - 900 \times 9.8 = 900 \times 0.4$ $T = 9180$ N

16 d)

Both $R(\uparrow)$: $T - 900 \times 9.8 = 900 \times -0.4$ $T = 8460$ N

16 e)

Both $R(\uparrow)$: $2700 - 900 \times 9.8 = 900 \times a$ $a = -6.8$ ms^{-2}

16 f)

Both $R(\uparrow)$: $8100 - 900 \times 9.8 = 900 \times a$ $a = -0.8$ ms^{-2}

16 g)

Both $R(\uparrow)$: $9450 - 900 \times 9.8 = 900 \times a$ $a = 0.7$ ms^{-2}

17 a)

$10 = 0 + \frac{1}{2}a \times 15^2 \Rightarrow a = \frac{4}{45}$

$a = 0.0889$ ms^{-2} (3 SF)

17 b)

Both R$(\uparrow)$: $T - 650 \times 10 = 650 \times \frac{4}{45}$

$\Rightarrow T = 6557.777\ldots$ so $T = 6560$ N (3 SF)

$T = 7000$ N (1 SF – AQA)

17 c)

R$(\uparrow)$: $R - 50 \times 10 = 50 \times \frac{4}{45} \Rightarrow R = \frac{4540}{9}$

$= 504.444\ldots$ so $R = 504$ N (3 SF)

$R = 500$ N (1 SF – AQA)

18 a)

All $R(\uparrow)$: $T - 115 \times 10 = 115 \times 0.4$ $T = 1196$ N 1000 N (1 SF – AQA)

18 b)

On P $R(\uparrow)$: $R - 5 \times 10 = 5 \times 0.4$ $R = 52$ N 50 N (1 SF – AQA)

18 c)

On lift $R(\uparrow)$: $1196 - 100 \times 10 - F = 100 \times 0.4$ $F = 156$ N 200 N (1 SF – AQA)

19 a)

All $R(\uparrow)$: $T - 138 \times 10 = 138 \times -0.25$ $T = 1345.5$ N 1000 N (1 SF – AQA)

19 b)

On P $R(\uparrow)$: $R - 3 \times 10 = 3 \times -0.25$ $R = 29.25$ N 30 N (1 SF – AQA)

19 c)

On lift $R(\uparrow)$: $1345.5 - 120 \times 10 - F = 120 \times -0.25$ $F = 175.5$ N 200 N (1 SF – AQA)

20 a)

All $R(\uparrow)$: $T - 126 \times 10 = 126 \times 0.1$ $T = 1272.6$ N 1000 N (1 SF – AQA)

20 b)

On lift $R(\uparrow)$: $1272.6 - 120 \times 10 - F = 120 \times 0.1$ $F = 60.6$ N 60 N (1 SF – AQA)

21 a)

$$a = \frac{dv}{dt} = -\frac{2}{5}t + 6$$
$$a|_{t=10} = 2 \text{ ms}^{-2}$$

21 b)

On Q: $\mathrm{R}(\rightarrow) : T = 5 \times 2 \Rightarrow T = 10$ N

21 b)

On P: $\mathrm{R}(\rightarrow) : D - 10 = 8 \times 2 \Rightarrow D = 26$ N